Classifications follow the ACMG/AMP Variant Interpretation Guidelines criteria (https://pubmed.ncbi.nlm.nih.gov/25741868/) with the following adjustments by the Sequence Variant Interpretation (SVI) Working Group. (Taken from: https://clinicalgenome.org/tools/clingen-variant-classification-guidance/ March 2026)

Updated recommendation for the benign stand-alone ACMG/AMP criterion

Ghosh R, Harrison SM, Rehm HL, Plon SE, Biesecker LG; ClinGen Sequence Variant Interpretation Working Group.

The Clinical Genome Resource (ClinGen) Sequence Variant Interpretation Working Group set out to refine the American College of Medical Genetics and Genomics and the Association of Molecular Pathologists (ACMG/AMP) variant pathogenicity recommendations for stand-alone rule BA1 (a variant with minor allele frequency [MAF] > 0.05 is benign), by clarifying how it should be used and specifying a set of variants that should be exempted from this rule. We cross-referenced ClinVar and Exome Aggregation Consortium data to identify variants for which there was a plausible argument for pathogenicity and the variant exists in one or more population data sets at MAF > 0.05. We identified nine such variants that were present in these data sets that may not be benign. The ACMG/AMP criteria were applied to these variants that resulted in four pathogenic and five variants of uncertain significance. We have refined benign rule BA1 by clarifying terms used to describe its use, which databases we recommend using, and assumptions made about this rule. We also recognized an initial list of nine variants for which there was some evidence of pathogenicity even though the MAF was high for these variants. We specify processes whereby individuals can petition ClinGen for amendments to our variant-specific assertions and the criteria experts should use when setting a numerically lower threshold for BA1 for specific genes.

Recommendations for interpreting the loss of function PVS1 ACMG/AMP variant criterion

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The 2015 ACMG/AMP sequence variant interpretation guideline provided a framework for classifying variants based on several benign and pathogenic evidence criteria, including a pathogenic criterion (PVS1) for predicted loss of function variants. However, the guideline did not elaborate on specific considerations for the different types of loss of function variants, nor did it provide decision-making pathways assimilating information about variant type, its location, or any additional evidence for the likelihood of a true null effect. Furthermore, this guideline did not take into account the relative strengths for each evidence type and the final outcome of their combinations with respect to PVS1 strength. Finally, criteria specifying the genes for which PVS1 can be applied are still missing. Here, as part of the ClinGen Sequence Variant Interpretation (SVI) Workgroup's goal of refining ACMG/AMP criteria, we provide recommendations for applying the PVS1 criterion using detailed guidance addressing the above-mentioned gaps. Evaluation of the refined criterion by seven disease-specific groups using heterogeneous types of loss of function variants (n = 56) showed 89% agreement with the new recommendation, while discrepancies in six variants (**11**%) were appropriately due to disease-specific refinements. Our recommendations will facilitate consistent and accurate interpretation of predicted loss of function variants.

Application of the ACMG/AMP Framework to Capture Evidence Relevant to Predicted and Observed Impact on Splicing: Recommendations from the ClinGen SVI Splicing Subgroup

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The American College of Medical Genetics and Genomics (ACMG) and the Association for Molecular Pathology (AMP) framework for classifying variants uses six evidence categories related to the splicing potential of variants: PVS1 (null variant in a gene where loss-of-function is the mechanism of disease), PS3 (functional assays show damaging effect on splicing), PP3 (computational evidence supports a splicing effect), BS3 (functional assays show no damaging effect on splicing), BP4 (computational evidence suggests no splicing impact), and BP7 (silent change with no predicted impact on splicing). However, the lack of guidance on how to apply such codes has contributed to variation in the specifications developed by different Clinical Genome Resource (ClinGen) Variant Curation Expert Panels. The ClinGen Sequence Variant Interpretation (SVI) Splicing Subgroup was established to refine recommendations for applying ACMG/AMP codes relating to splicing data and computational predictions. Our study utilised empirically derived splicing evidence to: 1) determine the evidence weighting of splicing-related data and appropriate criteria code selection for general use, 2) outline a process for integrating splicing-related considerations when developing a gene-specific PVS1 decision tree, and 3) exemplify methodology to calibrate bioinformatic splice prediction tools. We propose repurposing of the PVS1_Strength code to capture splicing assay data that provide experimental evidence for variants resulting in RNA transcript(s) with loss of function. Conversely BP7 may be used to capture RNA results demonstrating no impact on splicing for both intronic and synonymous variants, and for missense variants if protein functional impact has been excluded. Furthermore, we propose that the PS3 and BS3 codes are applied only for well-established assays that measure functional impact that is not directly captured by RNA splicing assays. We recommend the application of PS1 based on similarity of predicted RNA splicing effects for a variant under assessment in comparison to a known Pathogenic variant. The recommendations and approaches for consideration and evaluation of RNA assay evidence described aim to help standardise variant pathogenicity classification processes and result in greater consistency when interpreting splicing-based evidence.

SVI Recommendation for De Novo Criteria (PS2 & PM6) - Version 1.1

The Sequence Variant Interpretation (SVI) Working Group proposes a point -based system to determine the

strength of de novo evidence (ACMG/AMP criteria codes PS2 and PM6) based upon three parameters:

● confirmed parental relationships versus assumed parental relationships status

● phenotypic consistency

● number of de novo observations

To determine the appropriate strength level to apply for de novo occurrence(s), each proband with a de novo variant is awarded a point value based upon phenotypic consistency and confirmed or assumed parental relationships (Table 1). The combined point value of all de novo occurrences is then compared to Table 2 to determine the applicable evidence strength level. For example, if a NIPBL variant was de novo in one patient with Cornelia de Lange syndrome, with confirmed parental relationships (2 points; Table 1) and de novo in two additional unrelated patients with Cornelia de Lange syndrome with unconfirmed parental relationships (1 + 1 points; Table 1), then VeryStrong

evidence level is applied (PS2_VeryStrong) based on combined point value of 4 (Table 2). If the parents have not been tested for parentage or for the variant, no points should be awarded.

Table 1. Points* awarded per *de novo* occurrence

Phenotypic consistency	Points per Proband	
	de novo with confirmed parental relationships	*de novo* with unconfirmed parental relationships
Phenotype highly specific for gene	2	1
Phenotype consistent with gene but not highly specific	1	0.5
Phenotype consistent with gene but not highly specific and high genetic heterogeneity**	0.5	0.25
Phenotype not consistent with gene	0	0

*Note that these points are *not* equivalent to the points used to classify a variant per the Tavtigian et al 2020 "Fitting a naturally scaled point system to the ACMG/AMP variant classification guidelines"
**Maximum allowable value of 1 may contribute to overall score

Table 2. Recommendation for determining the appropriate ACMG/AMP evidence strength level for *de novo* occurrence(s)

Supporting (PS2_Supporting or PM6_Supporting)	Moderate (PS2_Moderate or PM6)	Strong (PS2 or PM6_Strong)	Very Strong (PS2_VeryStrong or PM6_VeryStrong)
0.5	1	2	4

For all uses of de novo criteria, the phenotype in the patient must be consistent with the gene/disease association as recommended in the ACMG/AMP guidelines. When the patient's phenotype is consistent with the gene/disease association but not highly specific, we recommend decreasing the points awarded. For example:

● A patient with early infantile epileptic encephalopathy and a de novo SIK1 variant with confirmed parental relationships is awarded 1 point (as the patient's phenotype is consistent with the gene but not highly specific and the variant is de novo with confirmed parental relationships). If this patient is the only de novo occurrence for the variant, then a Moderate strength level (PS2_Moderate) is applied.

○ If two additional unrelated patients with early infantile epileptic encephalopathy and a de novo SIK1 variant with confirmed parental relationships are identified, then the combined point value is 3 (as each patient is awarded 1 point). For these combined occurrences, a Strong strength level (PS2) is applied as the points reach the Strong threshold (2 points) but not the VeryStrong threshold (4 points).

● A patient with nonsyndromic intellectual disability and a de novo ASH1L variant is awarded 0.5 points (as the variant is de novo with confirmed parental relationships and patient's phenotype is consistent with the gene but not highly specific and there is significant evidence of genetic heterogeneity). If this patient is the only de novo occurrence for the variant, then a Supporting strength level (PS2_Supporting) is applied.

○ If a second patient with nonsyndromic intellectual disability and a de novo ASH1L variant with confirmed parental relationships is identified, then the combined point value is 1 (as each patient is awarded 0.5 points). For these combined occurrences, a Moderate strength level (PS2_Moderate) is applied.

● A patient with developmental delay but no other features of Cornelia de Lange syndrome and a de novo NIPBL variant with unconfirmed parental relationships is awarded zero points as this phenotype is not consistent with the gene/disease association. If this patient was the only de novo occurrence for the variant, then no de novo criteria are applied. Additional considerations for applying de novo criteria based on inheritance:

● Conditions with X-linked inheritance: if the variant occurs de novo in an unaffected carrier mother, and family history is consistent - i.e., she has no affected brothers/other male relatives apart from her affected son(s) – de novo criteria may be applied despite the fact that she is unaffected.

● Autosomal recessive conditions: for a de novo occurrence in a gene associated with a condition inherited in an autosomal recessive pattern without an additional pathogenic/likely pathogenic variant identified, the strength of evidence should be decreased by one level.

● Mosaicism: for cases with apparent germline mosaicism (multiple affected siblings with both parents negative for the variant), parental relationships must be confirmed in order for de novo criteria to apply.

Recommendations for application of the functional evidence PS3/BS3 criterion using the ACMG/AMP sequence variant interpretation framework

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Background The American College of Medical Genetics and Genomics (ACMG)/Association for Molecular Pathology (AMP) clinical variant interpretation guidelines established criteria (PS3/BS3) for functional assays that specified a “strong” level of evidence. However, they did not provide detailed guidance on how functional evidence should be evaluated, and differences in the application of the PS3/BS3 codes is a contributor to variant interpretation discordance between laboratories. This recommendation seeks to provide a more structured approach to the assessment of functional assays for variant interpretation and guidance on the use of various levels of strength based on assay validation.

Methods The Clinical Genome Resource (ClinGen) Sequence Variant Interpretation (SVI) Working Group used curated functional evidence from ClinGen Variant Curation Expert Panel-developed rule specifications and expert opinions to refine the PS3/BS3 criteria over multiple in-person and virtual meetings. We estimated odds of pathogenicity for assays using various numbers of variant controls to determine the minimum controls required to reach moderate level evidence. Feedback from the ClinGen Steering Committee and outside experts were incorporated into the recommendations at multiple stages of development.

Results The SVI Working Group developed recommendations for evaluators regarding the assessment of the clinical validity of functional data and a four-step provisional framework to determine the appropriate strength of evidence that can be applied in clinical variant interpretation. These steps

are: 1. Define the disease mechanism; 2. Evaluate applicability of general classes of assays used in the field; 3. Evaluate validity of specific instances of assays; 4. Apply evidence to individual variant interpretation. We found that a minimum of eleven total pathogenic and benign variant controls are required to reach moderate-level evidence in the absence of rigorous statistical analysis.

Conclusions The recommendations and approach to functional evidence evaluation described here should help clarify the clinical variant interpretation process for functional assays. Further, we hope that these recommendations will help develop productive partnerships with basic scientists who have developed functional assays that are useful for interrogating the function of a variety of genes.

SVI Recommendation for Absence/Rarity (PM2) - Version 1.0

The ClinGen Sequence Variant Interpretation (SVI) Working Group proposes decreasing the weight of criterion PM2 ("Absent from controls, or at extremely low frequency if recessive, in Exome Sequencing Project, 1000Genomes Project, or Exome Aggregation Consortium") from a Moderate strength level to a Supporting strength level (PM2_Supporting). After substantial analysis and modeling, the SVI WG proposes this weight adjustment due to concerns that absence or rarity is given too much weight in the 2015 framework and that this type of evidence does not meet the relative odds of pathogenicity for a Moderate pathogenic evidence, estimated to be 4.33:1 (PMID:29300386). This concern is supported by findings from the Exome Aggregation Consortium (ExAC) database that 99% of identified high-quality variants have a frequency <1%, that 54% of identified high-quality ExAC variants are only seen once in the entire data set, suggesting that rarity is actually common, and that all individuals harbor variants that are absent from the rest of the population (PMID: 27535533). By decreasing the weight of this criterion, this change will potentially impact variant classifications. Therefore the SVI proposes a novel criteria combination not listed in the combining rules outlined in the 2015 ACMG/AMP guideline. We propose that the combination of one Very Strong criterion and one Supporting criterion reach a classification of Likely pathogenic. This combination rule is supported by the Bayesian framework which shows that one Very Strong Pathogenic criterion and one Supporting Pathogenic criterion results in Post_P of 0.988, which falls within the Likely pathogenic range (0.90-0.99). This combining rule will allow novel LoF variants, that reach PVS1 and PM2 reduced to supporting, to still be classified as Likely pathogenic. We anticipate that other adjustments in the relative weight of evidence will be necessary in the future to accommodate this change in PM2 weight.

SVI Recommendation for in trans Criterion (PM3) - Version 1.0

The Sequence Variant Interpretation (SVI) Working Group proposes a point-based system to determine the strength of in trans observations (ACMG/AMP criterion PM3) based upon variant phasing and classification of the variant occurring on the other allele. Additionally, SVI recommends a revision to the criterion definition to indicate this evidence should only be applied if the individual is affected: SVI revision to PM3: For recessive disorders, detected in trans with a pathogenic or likely pathogenic variant in an affected patient To determine the appropriate strength level to apply for in trans occurrence(s), each proband is awarded a point value based upon phasing of the two variants in question (confirmed in trans versus unknown) and classification of the variant on the other allele (Table 1). The combined point value of all proband occurrences is then summed and compared to Table 2 to determine the applicable evidence strength level. For example, if assessing PAH variant NM_000277.3:c.1208C>T (p.Ala403Val) and the variant was confirmed in trans with Likely pathogenic variant c.1301C>A (p.Ala434Asp) in one proband (1.0 points; Table 1) and confirmed in trans with

Pathogenic variant c.331C>T (p.Arg111Ter) in another proband (1.0 points, Table 1), then PM3 at the Strong strength level (PM3_Strong) is applicable (2.0 points total; Table 2).

Table 1. Points awarded per in *trans* proband

Classification/Zygosity of other variant [1]	Points per Proband	
	Confirmed in *trans*	Phase unknown
Pathogenic or Likely pathogenic variant	1.0	0.5 (P) 0.25 (LP)
Homozygous occurrence *(max point 1.0)*	0.5	N/A
Uncertain significance variant *(max point 0.5)*	0.25	0.0

[1]All variants should be sufficiently rare (meet PM2 specification); P - Pathogenic; LP - Likely pathogenic

Table 2. Recommendation for determining the appropriate evidence strength level for PM3

PM3_Supporting	PM3	PM3_Strong	PM3_VeryStrong
0.5	1.0	2.0	4.0

Considerations:

● Allele Frequency - Application of PM3 is contingent on the allele frequency of the variant being assessed and the variant presumably on the other allele both being sufficiently rare (meets PM2 threshold). This contingency is to avoid incorrect application of PM3 to high frequency variants that are likely to occur in trans with P/LP variants based on frequency.

● Phasing - If the phase cannot be determined, it is recommended that at least two different LP/P variants (depending on classifications) are needed to equal the weight of one LP/P co-occurrence confirmed in trans.

○ In confirmation of phasing, if only one parent is tested and found to carry one allele, variants can be counted as in trans. For example, assessing PAH variant c.601C>T (p.His201Tyr) and variant was identified in PKU proband who also carries known pathogenic variant c.734T>C (p.Val245Ala). Only the mother is available for testing and the mother only carries c.734T>C (p.Val245Ala) variant, then variants can be considered in trans.

● Classification - Probands should be weighted less when the variant on the other allele is of uncertain significance and rare (meets PM2); however, weight may vary by gene size as larger genes are more likely to have a second variant by chance (default 0.25 points). If the variant on the other allele is classified as P or LP, weighting depends on phasing (see Phasing above), with P/LP being weighted equally if confirmed in trans and different point values per proband if phasing is unknown (0.5 points and 0.25 points, respectively). To avoid circularity, in all instances (phasing confirmed or unknown), the classification of the variant on the other allele should not use evidence from the variant being interrogated.

● Homozygous occurrences - For homozygous occurrences, the default weight is dropped to 0.5 points, as a rare homozygous occurrence may be due to consanguinity. A recommended max of 1.0

points of all homozygous cases is suggested to prevent overclassification of homozygous occurrences in the absence of additional data.

ClinGen guidance for use of the PP1/BS4 co-segregation and PP4 phenotype specificity criteria for sequence variant pathogenicity classification

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The 2015 American College of Medical Genetics and Genomics and the Association for Molecular Pathology variant classification publication established a standard employed internationally to guide laboratories in variant assessment. Those recommendations included both pathogenic (PP1) and benign (BS4) criteria for evaluating the inheritance patterns of variants, but details of how to apply those criteria at appropriate evidence levels were sparse. Several publications have since attempted to provide additional guidance, but anecdotally, this issue is still challenging. Additionally, it is not clear that those prior efforts fully distinguished disease-gene identification considerations from variant pathogenicity considerations nor did they address autosomal-recessive and X-linked inheritance. Here, we have taken a mixed inductive and deductive approach to this problem using real diseases as examples. We have developed a practical heuristic for genetic co-segregation evidence and have also determined that the specific phenotype criterion (PP4) is inseparably coupled to the co-segregation criterion. We have also determined that negative evidence at one locus constitutes positive evidence for other loci for disorders with locus heterogeneity. Finally, we provide a points-based system for evaluating phenotype and co-segregation as evidence types to support or refute a locus and show how that can be integrated into the Bayesian framework now used for variant classification and consistent with the 2015 guidelines.

Calibration of computational tools for missense variant pathogenicity classification and ClinGen recommendations for PP3/BP4 criteria

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Recommendations from the American College of Medical Genetics and Genomics and the Association for Molecular Pathology (ACMG/AMP) for interpreting sequence variants specify the use of computational predictors as "supporting" level of evidence for pathogenicity or benignity using criteria PP3 and BP4, respectively. However, score intervals defined by tool developers, and ACMG/AMP recommendations that require the consensus of multiple predictors, lack quantitative support. Previously, we described a probabilistic framework that quantified the strengths of evidence (supporting, moderate, strong, very strong) within ACMG/AMP recommendations. We have extended this framework to computational predictors and introduce a new standard that converts a tool's scores to PP3 and BP4 evidence strengths. Our approach is based on estimating the local positive predictive value and can calibrate any computational tool or other continuous-scale evidence on any variant type. We estimate thresholds (score intervals) corresponding to each strength of evidence for pathogenicity and benignity for thirteen missense variant interpretation tools, using carefully assembled independent data sets. Most tools achieved supporting evidence level for both pathogenic and benign classification using newly established thresholds. Multiple tools reached score thresholds justifying moderate and several reached strong evidence levels. One tool reached very strong evidence level for benign classification on some variants. Based on these findings, we provide

recommendations for evidence-based revisions of the PP3 and BP4 ACMG/AMP criteria using individual tools and future assessment of computational methods for clinical interpretation.

The ACMG/AMP reputable source criteria for the interpretation of sequence variants

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To the Editor: In 2015, the American College of Medical Genetics and Genomics and the Association for Molecular Pathology (ACMG/AMP) promulgated recommendations for the interpretation of sequence variants.[1] These recommendations have been widely implemented and shown to be useful for improving variant classification consistency.[2–4] From the beginning, they were recognized to be a starting point for further future refinements and extensions. The Clinical Genome (ClinGen) Resource is focused on curating the genome for use in molecular diagnosis.[5] One such effort is the Sequence Variant Interpretation Working Group, which has taken on the task of refining and evolving the current ACMG/AMP recommendations. This working group meets regularly and also, as individuals, interacts widely with the clinical testing community. Through these interactions, the working group has received input from multiple sources that two related criteria in the original recommendations should be considered for removal from the ACMG/AMP framework:

- PP5 “Reputable source recently reports variant as pathogenic, but the evidence is not available to the laboratory to perform an independent evaluation.”
- BP6 “Reputable source recently reports variant as benign, but the evidence is not available to the laboratory to perform an independent evaluation.”

It is our strongly held view that primary data are far preferable to expert opinion without access to primary data. The PP5 and BP6 criteria rely on assertions that are not directly linked to the evidence on which they were based. These criteria might have been appropriate in 2015 as they were originally intended as a bridge to allow the community to benefit from clinical laboratory expertise and locus-specific research databases, prior to the wider use of resources such as ClinVar that provide mechanisms for laboratories and researchers to share underlying primary data. Indeed, ClinVar has been even more successful than hoped and now includes 580,831 assertions for over 375,106 variants (ClinVar website accessed 8 January 2018). Submissions with “assertion criteria provided” review status, designating that the submitter either provided their evidence to ClinVar or indicated a willingness to provide evidence upon inquiry, account for 81% (470,245/580,831) of all submissions to ClinVar (ClinVar website accessed 8 January 2018). Therefore, there is less need to rely on assertions from reputable sources in the absence of primary evidence.

A second rationale for these two criteria was to support the efforts of the Sharing Clinical Reports Project (https://www.clinicalgenome.org/data-sharing/sharing-clinical-reports-project-scrp/), in which clinicians collected the test reports (including variant interpretation) produced by a large commercial laboratory that for the past decade has consistently declined to share underlying data or to submit assertions to ClinVar. As data for hereditary breast and ovarian cancer susceptibility alleles have increasingly been forthcoming from other laboratories, the necessity of this secondary information has declined and the currency of these data has receded.

Finally, we are concerned that these two criteria may be commonly misused by laboratories that incorporate primary data into variant assessment (e.g., functional data, criteria PS3 and BS3) and at the same time invoke criteria PP5 and BP6 for existing classifications that are based on the same set of data, which may lead to double counting, and potentially lead to errors in classification.

Based on these considerations, we propose that laboratories discontinue the use of criteria PP5 and BP6 as soon as that is practically achievable. We have removed these criteria from the ClinGen Variant Curation Interface. However, as with all types of evidence, interpretation of variants is the responsibility of the clinical testing laboratory director, who should account for the entirety of evidence and the sources of the data, and these recommendations should not be interpreted otherwise.